

3.6.1 Hardware

Phidgets 3-Axis Accelerometer

The accelerometer serves to give the amount of G force used to accelerate and corner when a vehicle is in operation. In addition, an accelerometer can provide useful information to derive other information such as horsepower of a vehicle, speed of vehicle and time to accelerate from 0-100km/h.

Phidgets 50A Ammeter

Depending on the kind of vehicle used, an electrical car is used for the prototype; hence an ammeter is used to measure the rate of use of power to propel the vehicle's movement. For a petrol/diesel vehicle, a flow meter will be used instead.

Phidgets Temperature Sensor

The temperature sensor is an essential probe to measure the environmental variable that might affect the driving conditions/fuel efficiency of a vehicle.

Electric Remote Control Car

A remote control car is used to emulate a real electrical car. The ammeter will be connected in series to the electrical motor circuits to measure amount of ampere used to move the vehicle forward.

Holux m1000 GPS Receiver

This GPS receiver is used to plot Confero's current location and map it on Microsoft Virtual Earth to provide navigational services. In addition, a GPS receiver can provide other useful information such as attitude, bearing and speed.

Wireless Modem

The wireless modem is to provide Internet access for Confero to call out services from Microsoft Virtual earth, such as route planning.

Kojinsha Touch Screen Ultra Mobile Personal Computer

The personal computer serves as the interface for all the probes to communication with our information system (Confero) to display and provide the user necessary information to operate their vehicle in the most efficient manner.